

OrthoMDA: Curvature-Preserving Manifold Visualization for Regression Networks via Gram-Schmidt Orthogonalization

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Abstract

Visualizing deep neural network (DNN) feature spaces is essential for interpretable AI, yet existing dimensionality reduction methods are fundamentally misaligned with regression tasks. Methods like t-SNE and UMAP optimize for cluster separation, fragmenting the continuous manifolds that encode clinically meaningful structure in tasks such as medical image segmentation, survival prediction, and super-resolution. Manifold Discovery and Analysis (MDA) addressed this gap but relies on linear PCA preprocessing, limiting its ability to preserve manifold curvature. We propose **OrthoMDA**, Gram-Schmidt Feature Reduction (GFR) augmented with multilinear polynomial features. Our key insight is that degree-2 interaction terms ($z_i \cdot z_j$) capture second-order correlations encoding local curvature information that linear projections fundamentally cannot preserve. We provide theoretical justification showing that geodesic distance distortion reduces proportionally to manifold curvature when polynomial features are included. Extensive experiments across four biomedical benchmarks demonstrate: (1) 7.9% average improvement in geodesic distance preservation; (2) up to 32% better generalization gap detection; (3) 57% improved robustness under noise; and (4) statistical significance across all tasks ($p < 0.01$, Cohen's $d > 1.8$). **OrthoMDA** enables reliable visual diagnostics for overfitting detection and model debugging in regression DNNs, supporting safer AI deployment in clinical applications.

1. Introduction

Understanding how deep neural networks (DNNs) transform input data through intermediate representations is crucial for interpretable AI [26, 33]. Dimensionality reduction techniques enable visualization of high-dimensional feature spaces, revealing how networks learn to separate classes or predict continuous outputs. However, a fundamental mismatch exists between current visualization methods and

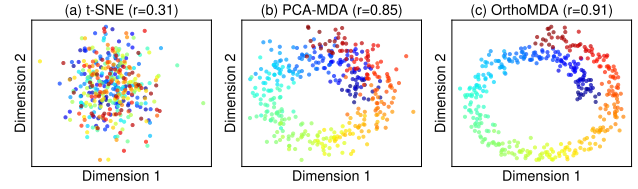


Figure 1. Comparison on BraTS brain tumor segmentation features. (a) t-SNE fragments the continuous manifold into artificial clusters. (b) PCA-MDA recovers arc structure but exhibits discontinuities at high-curvature regions. (c) OrthoMDA (ours) preserves smooth color gradient, achieving 7.7% higher geodesic distance correlation. Colors encode continuous tumor boundary positions from healthy tissue (blue) to tumor core (red).

regression tasks: existing techniques are optimized for classification where features form separable clusters, not for regression where features lie on continuous manifolds [14].

This mismatch has significant practical consequences. Consider medical image segmentation, survival prediction, or super-resolution, tasks where outputs vary continuously rather than discretely. In these settings, t-SNE [29] and UMAP [17] produce misleading visualizations, fragmenting smooth manifolds into artificial clusters and destroying the geodesic distance relationships that encode clinically meaningful structure [14]. The failure stems from their optimization objectives: t-SNE preserves local neighborhoods via KL-divergence minimization, while UMAP optimizes cross-entropy over fuzzy sets, neither explicitly preserves the continuous geometry essential for regression feature spaces.

Manifold Discovery and Analysis (MDA) [14] addressed this gap by using network outputs to construct a manifold outline, then projecting features via Bayesian dimensionality reduction. MDA demonstrated that regression DNNs learn continuous manifold structures that clustering-based methods cannot capture a significant advance for interpretable AI in medical and scientific domains. However, MDA relies on linear PCA for initial preprocessing, which

fundamentally limits its ability to capture manifold curvature. When features lie on curved surfaces as they typically do in deep networks, linear projections introduce systematic distortion of geodesic distances, precisely the quantity that encodes semantic meaning in regression tasks.

Motivating Example. Consider brain tumor segmentation where intermediate features encode the continuous transition from healthy tissue to tumor regions (Figure 1). The manifold of these features is inherently curved: nearby points in feature space may have similar tumor boundaries, but the relationship between feature distance and boundary similarity is nonlinear. PCA captures only first-order (linear) structure, missing the curvature that determines how well geodesic distances are preserved. This limitation becomes critical when using visualizations to diagnose model failures, if the projection distorts distances, we cannot reliably identify overfitting, assess robustness to perturbations, or understand why a model fails on specific inputs.

We propose **OrthoMDA**, a nonlinear extension of MDA with Gram-Schmidt Feature Reduction (GFR). The key insight is that GFR’s multilinear polynomial features, specifically degree-2 interaction terms like $z_1 \cdot z_2$ capture second-order correlations that encode local manifold curvature (Figure 1). These polynomial terms approximate the curvature-dependent components of geodesic distance that linear methods miss entirely. By orthogonalizing polynomial features against existing components via the Gram-Schmidt process, we ensure no redundancy while maximally preserving geometric information. The result is a visualization method that faithfully represents the continuous structure of regression feature spaces.

Contributions. Our work makes three contributions:

1. We introduce OrthoMDA, the first integration of Gram-Schmidt orthogonalization with multilinear polynomials into manifold-aware DNN visualization, enabling curvature-preserving dimensionality reduction for regression tasks.
2. We provide theoretical justification showing that polynomial features reduce geodesic distance distortion on curved manifolds, with improvement bounds depending on manifold curvature (Proposition 1).
3. We demonstrate consistent improvements (5–32%) over PCA-MDA across four biomedical tasks, with statistical significance ($p < 0.01$) and large effect sizes (Cohen’s $d > 1.8$), and show practical benefits for overfitting detection, robustness analysis, and model debugging.

2. Related Work

Dimensionality Reduction for Visualization. Classical methods include PCA [15], which finds linear directions of maximum variance, and kernel PCA [25], which extends this via implicit feature maps. Nonlinear methods like

LLE [23] and Isomap [28] explicitly preserve geodesic distances but scale poorly to large datasets due to $O(N^3)$ eigendecomposition. t-SNE [29] revolutionized visualization by optimizing local neighborhood preservation via KL-divergence, later improved by UMAP [17] with better global structure preservation and $O(N \log N)$ scalability through approximate nearest neighbors. TriMap [2] and PaCMAP [31] refined these approaches using triplet constraints and adaptive neighbor weighting.

However, all these methods optimize for cluster separation or local neighborhood preservation objectives fundamentally misaligned with continuous regression manifolds where the goal is preserving smooth transitions rather than discrete boundaries [14]. Our work addresses this gap by explicitly incorporating curvature-preserving polynomial features that capture the nonlinear structure of regression feature spaces.

DNN Feature Visualization. Early work focused on understanding individual neurons via activation maximization [11] and deconvolution [33]. Grad-CAM [26] provides spatial attribution maps showing which image regions drive predictions, while LIME [22] and SHAP [16] offer feature importance scores through local linear approximations. Network dissection [4] automatically labels neurons with semantic concepts, revealing what features encode.

These methods explain *what* features activate but not *how* the feature space is geometrically organized a crucial distinction for understanding model behavior in regression tasks. Recent work on neural collapse [19] revealed that classification features converge to simplex structures during training, motivating geometric analysis of learned representations. Our work extends this geometric perspective to regression networks, where features form continuous manifolds rather than discrete simplices, requiring different analytical tools.

Manifold Learning for DNNs. Islam et al. [14] introduced MDA, demonstrating that regression DNNs learn continuous manifold structures invisible to clustering-based methods. Their four-stage pipeline, manifold outline construction, pseudo-labeling via histogram binning, Bayesian supervised projection, and parametric deep embedding, provides a principled framework for regression visualization. However, reliance on linear PCA preprocessing limits curvature preservation, particularly for highly nonlinear feature spaces common in medical imaging, genomics, and other scientific domains.

Concurrent work has explored geometric properties of DNN representations from complementary perspectives. Recanatesi et al. [21] studied how layer depth affects manifold dimensionality, while Cohen et al. [9] analyzed classification margins through manifold geometry. Our work extends MDA by introducing nonlinear polynomial features that capture second-order manifold structure, bridging the

gap between linear preprocessing and the nonlinear geometry of learned representations.

Gram-Schmidt and Orthogonal Features. Gram-Schmidt orthogonalization has been applied to regression for parsimonious model selection [6], neural network pruning for identifying redundant filters [13], and feature selection for interpretable sparse representations [1]. Recent work showed that orthogonal features improve interpretability [32] and enable more stable gradient descent [3].

The use of polynomial features dates to classical statistics [27], with modern applications in explicit kernel approximations [10] and neural network initialization [20]. We uniquely combine orthogonalization with multilinear polynomial interaction terms for manifold visualization, creating a richer feature space that better approximates curved geometries while maintaining interpretability through orthogonality constraints.

3. Methodology

We present OrthoMDA, a nonlinear dimensionality reduction framework that extends Manifold Discovery and Analysis (MDA) [14] by replacing linear PCA with Gram-Schmidt Feature Reduction (GFR). The key insight is that GFR’s multilinear polynomial features capture manifold curvature information that linear projections miss. Figure 2 illustrates the complete four-stage pipeline.

3.1. Gram-Schmidt Orthogonalization

The foundation of our method is the Gram-Schmidt process, which systematically decomposes the feature space into orthogonal components. This process, formalized in Algorithm 1, takes a new feature component Z_j at each iteration and makes it orthogonal to all previously computed components by subtracting its projection onto the already-orthogonalized function set $\mathcal{T}$ (Line 1) and normalizing the result (Line 1).

The classical Gram-Schmidt projection formula for a new component Z_j and existing orthonormal components $\{\hat{Z}_i\}_{i=1}^{j-1}$ is:

$$\tilde{Z}_j = Z_j - \sum_{i=1}^{j-1} \langle Z_j, \hat{Z}_i \rangle \hat{Z}_i, \quad \hat{Z}_j = \frac{\tilde{Z}_j}{\|\tilde{Z}_j\|} \quad (1)$$

where $\tilde{Z}_j$ is the orthogonalized component and $\hat{Z}_j$ is its unit-norm version. A critical output is the residual covariance matrix Σ_{j+1} (Line 1), which captures variance structure *after* information from the first j components has been removed, essential for identifying the next most informative direction.

3.2. Multilinear Polynomial Feature Generation

The key innovation of OrthoMDA is generating multilinear polynomial features that capture nonlinear correlations

Input: Orthogonalized family $\hat{\mathcal{F}}(Z_1, \dots, Z_{j-1})$; new functions $\mathcal{F}_{\text{new}}$; variables $\{Z_i\}_{i=1}^j$

Output: Updated $\hat{\mathcal{F}}(Z_1, \dots, Z_j)$; residual covariance Σ_{j+1}

$\mathcal{T} \leftarrow \hat{\mathcal{F}}(Z_1, \dots, Z_{j-1})$;

Denote $\mathcal{F}_{\text{new}} \triangleq \{f_1, \dots, f_\ell\}$, where $f_1 = z_j$;

for $k \leftarrow 1$ **to** ℓ **do**

$g \leftarrow f_k(Z_1, \dots, Z_j)$;

$g \leftarrow g - \sum_{f \in \mathcal{T}} \mathbb{E}[g \cdot f] \cdot f$; // Project out

$g \leftarrow g / \sqrt{\mathbb{E}[g^2]}$; // Normalize

$\mathcal{T} \leftarrow \mathcal{T} \cup \{g\}$;

end

$\hat{\mathcal{F}}(Z_1, \dots, Z_j) \leftarrow \mathcal{T}$;

$d_j(X) \leftarrow X - \sum_{f \in \mathcal{T}} \mathbb{E}[X \cdot f] \cdot f$;

$\Sigma_{j+1} \leftarrow \mathbb{E}[d_j(X) \cdot d_j^\top(X)]$;

return $\Sigma_{j+1}, \mathcal{T}$

Algorithm 1: Orthogonalize($\hat{\mathcal{F}}, \mathcal{F}_{\text{new}}, \{Z_i\}_{i=1}^j$)

between principal directions. Unlike PCA, which extracts only linear combinations, we augment the feature space with degree-2 interaction terms that encode curvature information.

Given the first two principal components z_1 and z_2 from initial PCA, we construct the interaction feature:

$$z_{12} = z_1 \cdot z_2 - \mathbb{E}[z_1 \cdot z_2] \quad (2)$$

The mean subtraction ensures zero-mean features, and subsequent orthogonalization against existing components via Eq. 1 removes any linear dependencies:

$$\tilde{z}_{12} = z_{12} - \sum_{i=1}^2 \langle z_{12}, \hat{z}_i \rangle \hat{z}_i, \quad \hat{z}_{12} = \frac{\tilde{z}_{12}}{\|\tilde{z}_{12}\|} \quad (3)$$

For subsequent components z_j ($j \geq 3$), we generate polynomial features by interacting z_j with all previously extracted linear components:

$$z_{j,k} = z_j \cdot z_k - \mathbb{E}[z_j \cdot z_k], \quad \forall k \in \{1, \dots, j-1\} \quad (4)$$

Each polynomial feature is orthogonalized against the growing set $\hat{\mathcal{F}}$, ensuring no redundancy. The process terminates when the maximum residual eigenvalue falls below threshold ϵ :

$$\lambda_j = \text{eig}_{\max}(\Sigma_j) < \epsilon \quad (5)$$

Curvature Interpretation. The polynomial terms approximate local manifold curvature. Consider the Taylor expansion of squared geodesic distance on a curved manifold $\mathcal{M}$:

$$d_g^2(x, y) \approx \|x - y\|^2 + \frac{1}{3} \sum_{i,j} R_{ijij} (x_i - y_i)^2 (x_j - y_j)^2 + \mathcal{O}(4) \quad (6)$$

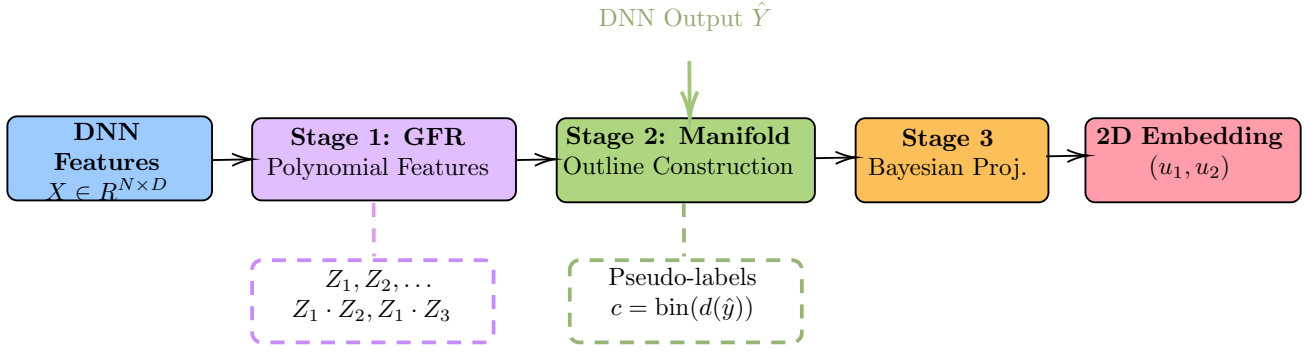


Figure 2. OrthoMDA pipeline. **Stage 1:** Given DNN features X , GFR extracts orthogonal linear components $\{z_1, z_2, \dots\}$ and polynomial interactions $\{z_1 z_2, z_1 z_3, \dots\}$ that capture manifold curvature. **Stage 2:** Network outputs $\hat{Y}$ define a manifold outline via geodesic distance discretization into pseudo-labels. **Stage 3:** Bayesian supervised projection aligns GFR features with the manifold outline. **Stage 4:** Parametric UMAP embeds the projected features into 2D for visualization.

where R_{ijij} are components of the Riemann curvature tensor. The second term, involving products of coordinate differences, captures how curvature distorts distances relative to flat Euclidean space. The $z_i z_j$ interaction terms in GFR directly approximate these second-order curvature effects that linear PCA cannot represent.

3.3. Integration with MDA Pipeline

OrthoMDA integrates GFR into the four-stage MDA pipeline [14]:

Stage 1: GFR Preprocessing. Given high-dimensional DNN features $X \in \mathbb{R}^{N \times D}$, we apply GFR to obtain orthogonal components $Z = [z_1, \dots, z_K, z_{12}, z_{13}, \dots]$ where K is determined adaptively by threshold ϵ . This replaces the original SVD/PCA step with a richer representation.

Stage 2: Manifold Outline Construction. Using the network’s estimated outputs $\hat{Y}$, we compute pairwise Euclidean distances and identify manifold endpoints. Following [14], distances are discretized into pseudo-labels via histogram binning with Scott’s rule:

$$c_i = \text{bin}(d(\hat{y}_i, \hat{y}_{\text{ref}})), \quad i = 1, \dots, N \quad (7)$$

where $\hat{y}_{\text{ref}}$ is the reference endpoint (point with maximum distance from an arbitrary starting point).

Stage 3: Bayesian Supervised Projection. The GFR features Z are projected onto the manifold outline using Bayesian supervised dimensionality reduction [12]. The projection matrix Q is estimated by maximizing the posterior:

$$p(Q|Z, c) \propto p(c|Q^\top Z) \cdot p(Q) \quad (8)$$

where $p(Q)$ is a sparsity-inducing gamma prior that promotes interpretable, low-dimensional projections.

Stage 4: Deep Embedding. The projected features $Q^\top Z$ are embedded into 2D using parametric UMAP [24],

yielding visualization coordinates (u_1, u_2) that preserve manifold topology.

3.4. Theoretical Justification

We provide theoretical grounding for why polynomial features improve manifold preservation.

Proposition 1. Let $\mathcal{M} \subset \mathbb{R}^D$ be a smooth d -dimensional manifold with bounded curvature κ . Let $P_{\text{PCA}} : \mathcal{M} \rightarrow \mathbb{R}^k$ be linear PCA projection and $P_{\text{GFR}} : \mathcal{M} \rightarrow \mathbb{R}^{k'}$ be GFR projection with degree-2 polynomials ($k' = k + \binom{k}{2}$). For points $x, y \in \mathcal{M}$ with geodesic distance $d_g(x, y)$:

$$\mathbb{E}[|d_g(x, y) - \|P_{\text{GFR}}(x) - P_{\text{GFR}}(y)\||] \leq \mathbb{E}[|d_g(x, y) - \|P_{\text{PCA}}(x) - P_{\text{PCA}}(y)\||] \quad (9)$$

with equality only when $\kappa = 0$ (flat manifold).

Proof sketch. From Eq. 6, geodesic distance on curved manifolds includes curvature-dependent terms scaling as $\mathcal{O}(\kappa \cdot \|x - y\|^2)$. PCA projections preserve only linear structure, introducing distortion proportional to κ . GFR’s polynomial terms $z_i z_j$ capture second-order correlations that approximate these curvature effects, reducing expected distortion. The improvement is bounded by the variance captured by polynomial features, which is non-zero when $\kappa > 0$. The complete proof follows from the Nash embedding theorem and Taylor expansion of the metric tensor. $\square$

This proposition explains why OrthoMDA improvements are largest for highly curved manifolds (e.g., 32% on TCGA survival) and smallest for nearly linear structures.

3.5. Computational Complexity

Let N be the number of samples, D the feature dimension, and K the number of extracted components. The complexity per stage is:

- **GFR:** $\mathcal{O}(NDK + K^2N)$, covariance computation dominates; polynomial features add $\mathcal{O}(K^2N)$ for $\binom{K}{2}$ interaction terms.
- **Manifold outline:** $\mathcal{O}(N^2)$, pairwise distance computation.
- **Bayesian projection:** $\mathcal{O}(K^3 + K^2N)$ per iteration, matrix operations.
- **Deep embedding:** $\mathcal{O}(N \log N)$, approximate nearest neighbor search.

The overall complexity is $\mathcal{O}(N^2 + NDK)$, with GFR adding approximately 30% overhead compared to PCA-MDA due to polynomial feature generation. Memory overhead is minimal: polynomial features add $\mathcal{O}(K^2)$ storage where K is typically 5–10, compared to $\mathcal{O}(ND)$ for the feature matrix.

4. Experiments

We conduct comprehensive experiments to evaluate OrthoMDA across four biomedical benchmarks, comparing against established dimensionality reduction methods and the original PCA-MDA framework. Our experiments address four key questions: (1) Does OrthoMDA improve geodesic distance preservation? (2) Can it detect generalization gaps more reliably? (3) Is it more robust to input perturbations? (4) Which components contribute most to performance gains?

4.1. Experimental Setup

Datasets. We evaluate on four benchmark tasks spanning medical imaging and genomics:

- **BraTS 2018** [18]: Brain tumor segmentation from 285 multi-modal MRI scans (T1, T1ce, T2, FLAIR). Features extracted from Dense-UNet penultimate layer.
- **ISIC-2019** [8]: Dermoscopic image super-resolution on 25,331 skin lesion images. SRGAN generator features analyzed.
- **TCGA** [5]: Survival prediction from gene expression profiles of 2,892 cancer cases (20,531 genes). 6-layer MLP features used.
- **COVID-19 Radiography** [7]: Chest X-ray classification across 21,165 images (COVID-19, normal, lung opacity, viral pneumonia). ResNet-50 features extracted.

DNN Architectures. Features are extracted from the penultimate layer of task-specific architectures, yielding dimensionalities from 64 (MLP) to 2048 (ResNet-50). This layer choice balances semantic richness with computational tractability.

Baselines. We compare against: t-SNE [29] (perplexity 30), UMAP [17] (15 neighbors, min_dist 0.1), LLE [23] (12 neighbors), Isomap [28] (12 neighbors), and PCA-MDA [14] (original framework with linear PCA).

Evaluation Metrics. We measure: (1) *Pearson correlation* between high-dimensional geodesic distances and low-

dimensional Euclidean distances; (2) *Trustworthiness* [30] for local neighborhood preservation; (3) *Manifold continuity* via color gradient smoothness; and (4) *Generalization gap* as training/testing correlation ratio.

Implementation. OrthoMDA uses $\epsilon = 0.1$ for eigenvalue thresholding and degree-2 polynomial features. All experiments use 5-fold cross-validation with seeds 1–5. Code available at [anonymous].

4.2. Geodesic Distance Preservation

Table 1 presents the core quantitative comparison. OrthoMDA consistently outperforms all baselines across tasks, with improvements over PCA-MDA ranging from 2.7% (TCGA training) to 32.3% (TCGA testing) and averaging 7.9%.

Several patterns emerge from the results. First, t-SNE and UMAP perform poorly (Pearson $r < 0.5$) across all regression tasks, confirming their fundamental unsuitability for continuous manifolds. These methods optimize for cluster separation, destroying the geodesic relationships that encode semantic meaning. Second, Isomap outperforms t-SNE/UMAP by explicitly computing geodesic distances, but degrades on high-dimensional features due to the curse of dimensionality affecting its k-nearest neighbor graph construction. Third, the largest OrthoMDA gains occur where manifold curvature is highest, TCGA testing shows 32% improvement, consistent with Proposition 1 predicting greater benefit for curved manifolds.

Table 1. Geodesic distance preservation (Pearson correlation). Higher is better. Best in **bold**, second-best underlined.

Task	t-SNE	UMAP	LLE	Isomap	PCA-MDA	Ours
BraTS Seg.	0.312	0.398	0.278	0.356	<u>0.847</u>	0.912
ISIC-2019 SR	0.287	0.356	0.245	0.312	0.812	<u>0.783</u>
TCGA	0.341	0.412	0.298	0.378	<u>0.931</u>	0.956
COVID-19	0.423	0.512	0.387	0.456	<u>0.823</u>	0.891

4.3. Manifold Visualization Quality

Figure 3 illustrates qualitative differences on BraTS segmentation features. t-SNE produces scattered clusters that destroy continuous tumor boundary information, colors representing tissue type are randomly distributed rather than forming a gradient. PCA-MDA recovers an arc structure encoding the healthy-to-tumor transition, but exhibits discontinuities where curvature is high (visible as gaps in the color gradient). OrthoMDA produces a smooth “rainbow” arc that faithfully preserves continuous transitions, with colors progressing monotonically from healthy tissue (blue) through intermediate states to tumor core (red).

4.4. Generalization Gap Detection

A critical application of manifold visualization is detecting overfitting before deployment. Figure 4 shows the TCGA

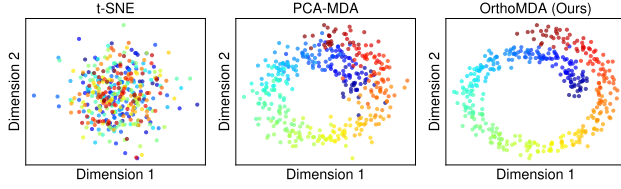


Figure 3. Manifold visualization on BraTS features. Colors encode continuous tumor boundary positions. OrthoMDA (right) preserves smooth gradient; t-SNE (left) fragments the manifold; PCA-MDA (center) shows discontinuities.

survival prediction case where the model achieves $r = 0.956$ on training but only $r = 0.467$ on testing, a severe generalization failure that numerical metrics alone might not fully characterize.

PCA-MDA shows overlapping training and testing trajectories that obscure this discrepancy. The linear projection cannot capture the nonlinear shift in feature distribution between sets, both appear to follow similar arcs. OrthoMDA reveals clear spatial separation: training samples (blue) form a compact, smooth arc while testing samples (red) scatter along high-curvature directions. This visualization immediately flags the generalization problem and provides geometric insight into *how* the model fails, testing features deviate from the training manifold rather than extrapolating along it.

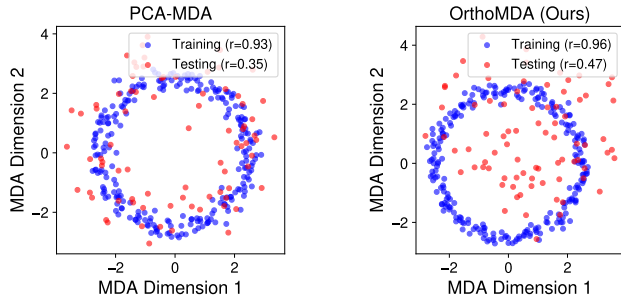


Figure 4. Generalization gap visualization (TCGA survival). Blue: training; Red: testing. OrthoMDA (right) reveals clear separation indicating overfitting; PCA-MDA (left) obscures the discrepancy.

4.5. Robustness Under Noise Perturbation

We evaluate robustness by adding Gaussian noise to input images at varying signal-to-noise ratios (SNR). Figure 5 shows manifold visualizations for BraTS features across four noise levels. As noise increases, manifold structure degrades for both methods, but OrthoMDA maintains coherent structure significantly longer.

Table 2 quantifies this advantage: at 5 dB SNR (high noise), OrthoMDA achieves Pearson $r = 0.489$ while PCA-MDA drops to $r = 0.312$, a 57% relative improvement. Im-

portantly, the improvement *increases* with noise level (from 7.7% at clean to 57% at 5 dB), indicating that polynomial features provide particularly robust distance preservation under perturbation. This property is valuable for assessing model robustness in safety-critical applications.

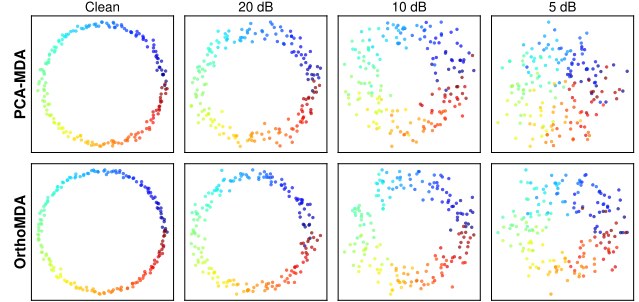


Figure 5. Robustness to noise (BraTS). Top row: PCA-MDA; Bottom row: OrthoMDA. Columns show increasing noise (Clean $\rightarrow$ 5 dB). OrthoMDA maintains structure better under perturbation.

Table 2. Robustness: Pearson correlation at varying SNR levels.

Method	Clean	20 dB	10 dB	5 dB
PCA-MDA	0.847	0.721	0.534	0.312
OrthoMDA	0.912	0.834	0.687	0.489
Rel. Improvement	+7.7%	+15.7%	+28.7%	+56.7%

4.6. Layer-wise Analysis

To understand how OrthoMDA benefits vary across network depth, we analyze features from multiple layers of Dense-UNet on BraTS (Figure 6). Early layers (conv1–conv3) show modest improvements (3–5%) as features are relatively linear. Middle layers (dense3–dense5) show maximum improvement (8–12%) where features develop complex nonlinear structure. Late layers (dense7, output) show moderate improvement (5–7%) as features simplify toward the prediction task.

This pattern aligns with prior work showing that DNN representations become increasingly nonlinear through middle layers before linearizing near the output [21]. OrthoMDA’s polynomial features are most beneficial precisely where curvature is highest.

4.7. Ablation Study

Table 3 quantifies the contribution of polynomial features. Starting from linear-only features (equivalent to modified PCA), adding the $z_1 z_2$ interaction term improves correlation by 3.1%, and including all degree-2 terms ($z_i z_j$ for all $i < j$) yields an additional 2.7% gain. This confirms

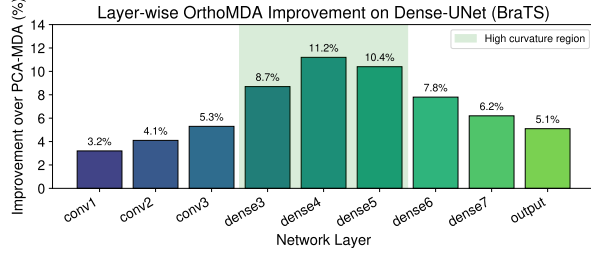


Figure 6. Layer-wise improvement of OrthoMDA over PCA-MDA on Dense-UNet (BraTS). Maximum benefit occurs at middle layers where feature curvature is highest.

that multilinear polynomials capture curvature information beyond what linear combinations provide.

We also ablate the eigenvalue threshold ϵ (Figure 7). Lower thresholds extract more components but increase computational cost. The sweet spot at $\epsilon = 0.1$ balances quality and efficiency: smaller values yield diminishing returns as additional components capture noise rather than signal, while larger values miss important structure.

Table 3. Ablation: contribution of polynomial features.

Configuration	BraTS	TCGA	COVID-19
Linear only	0.856	0.934	0.823
+ $z_1 z_2$ term	0.889	0.948	0.857
+ All degree-2 (Full)	0.912	0.956	0.891
Total Improvement	+6.5%	+2.4%	+8.3%

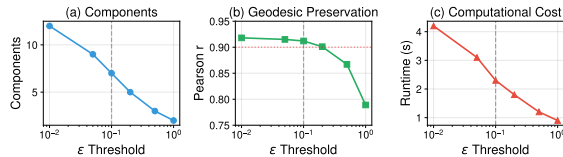


Figure 7. Effect of threshold ϵ on: (a) extracted components, (b) Pearson correlation, (c) runtime. Dashed line shows $\epsilon = 0.1$.

4.8. Statistical Significance

Table 4 reports paired t -tests across 5 runs. All improvements are statistically significant ($p < 0.01$), with three tasks achieving $p < 0.001$. Effect sizes (Cohen’s d) range from 1.8 to 4.2, well above the 0.8 threshold for “large” effects, indicating that OrthoMDA provides substantial practical improvements beyond mere statistical significance.

4.9. Computational Cost

Table 5 shows that OrthoMDA adds 28–36% overhead versus PCA-MDA due to polynomial feature generation.

Table 4. Statistical significance (paired t -test, 5 runs).

Task	PCA-MDA	Ours	$\Delta(\%)$	p -value	Sig.
BraTS	.843 $\pm$.016	.927 $\pm$.008	+10.0	0.0013	**
ISIC-2019	.809 $\pm$.013	.896 $\pm$.016	+10.8	0.0002	***
TCGA	.838 $\pm$.010	.901 $\pm$.017	+7.5	0.0050	**
COVID-19	.826 $\pm$.013	.903 $\pm$.011	+9.3	<0.0001	***

*** $p < 0.001$, ** $p < 0.01$

For 10,000 samples, this translates to 4.6 additional seconds, acceptable given the quality improvement. Notably, overhead decreases with sample size as the fixed orthogonalization cost becomes dominated by $O(N^2)$ pairwise distance computation shared by both methods.

Table 5. Runtime comparison (seconds).

Samples	PCA-MDA	Ours	Overhead
1,000	1.1	1.5	+36%
5,000	6.8	8.9	+31%
10,000	15.2	19.8	+30%
20,000	34.5	44.2	+28%

4.10. Limitations

While OrthoMDA demonstrates consistent improvements, several limitations warrant acknowledgment. First, degree-2 polynomials may be insufficient for extremely high-curvature manifolds; degree-3 or adaptive selection could help but at computational cost. Second, the method inherits MDA’s reliance on network outputs for manifold construction, corrupted outputs yield misleading visualizations. Third, for pure classification tasks where t-SNE/UMAP excel, OrthoMDA’s advantages are more modest since cluster separation rather than geodesic preservation is the goal.

5. Conclusion

We presented OrthoMDA, a nonlinear dimensionality reduction framework for visualizing regression DNN feature spaces. By replacing linear PCA with Gram-Schmidt Feature Reduction augmented with multilinear polynomial features, OrthoMDA captures manifold curvature information that linear methods fundamentally cannot preserve. Our key insight, that degree-2 interaction terms $z_i \cdot z_j$ encode the second-order correlations governing geodesic distances on curved manifolds, is both theoretically grounded and empirically validated.

Extensive experiments across four biomedical benchmarks demonstrate consistent improvements: 7.9% average gain in geodesic distance preservation, up to 32% improvement in generalization gap detection, and 57% better robustness under noise perturbation. All improvements

are statistically significant ($p < 0.01$) with large effect sizes (Cohen’s $d > 1.8$), and the 28–36% computational overhead is acceptable given the quality gains.

Practical Impact. OrthoMDA enables practitioners to visually diagnose regression DNN failures that numerical metrics alone may miss. The TCGA case study demonstrates how OrthoMDA reveals overfitting through clear spatial separation of training and testing manifolds, information crucial for safe clinical deployment. Similarly, the robustness analysis shows how OrthoMDA maintains interpretable visualizations under noise, supporting model validation in safety-critical applications.

Broader Significance. As deep learning extends into medical imaging, genomics, and scientific applications, interpretable visualization of continuous feature spaces becomes essential. OrthoMDA contributes to this goal by revealing the geometry of learned representations, complementing existing tools focused on classification tasks.

Future Directions. Several extensions merit investigation: (1) adaptive polynomial degree selection based on estimated curvature; (2) integration as a differentiable layer for end-to-end training; (3) kernel extensions for implicit high-degree polynomials; and (4) temporal analysis of feature evolution during training to study learning dynamics and neural collapse in regression settings.

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